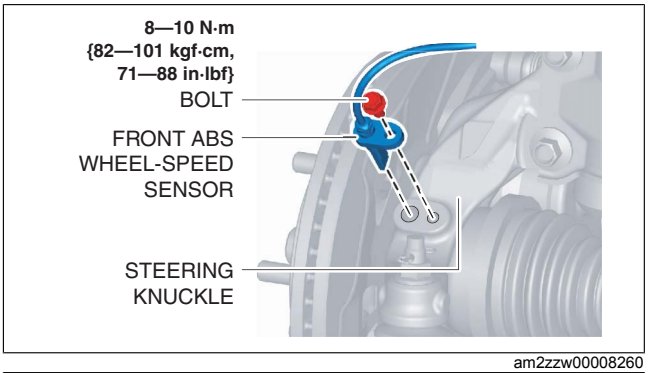
FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

id0213008010f2

Caution

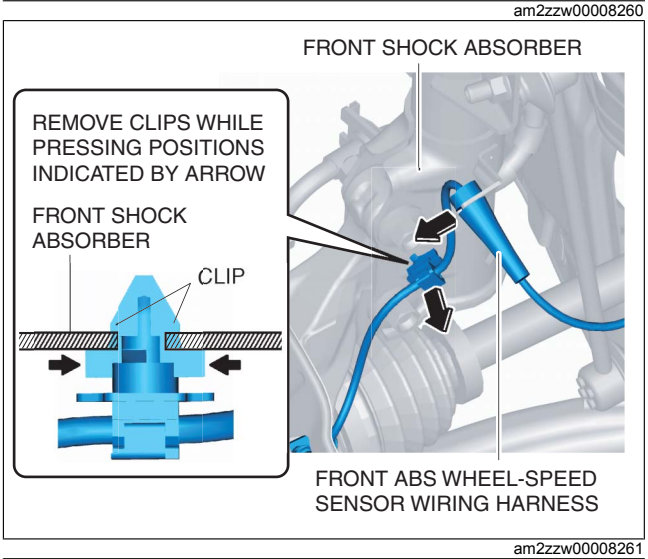
- **Performing the following procedures without first removing the front ABS wheel-speed sensor may possibly cause an open circuit in the harness if it is pulled by mistake. Before performing the following procedures, disconnect the front ABS wheel-speed sensor (axle side) and fix it to an appropriate place where the sensor will not be pulled by mistake while servicing the vehicle.**
- **Secure the steering wheel using tape or a cable to prevent the steering shaft from rotating after disconnecting the steering shaft. If the steering wheel rotates after the steering shaft and the steering gear and linkage are disconnected, the internal parts of the clock spring could be damaged.**

1. Remove the joint cover. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
2. Disconnect the intermediate shaft from the steering gear and linkage. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
3. Remove the front ABS wheel-speed sensor from the steering knuckle.



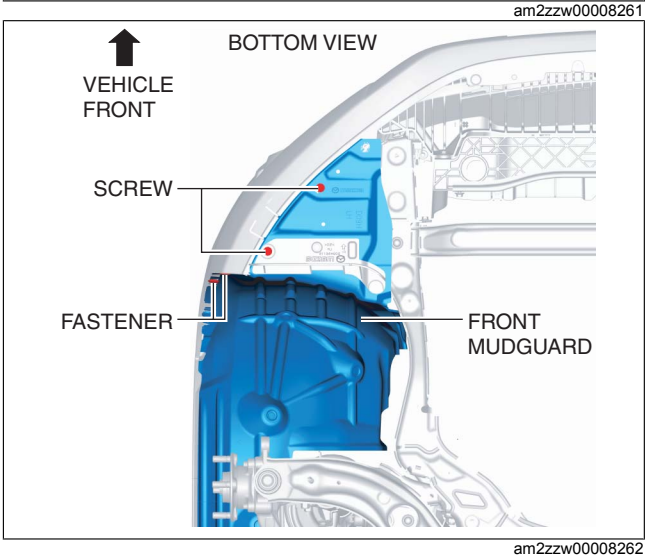
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4. Remove the front ABS wheel-speed sensor wiring harness from the front shock absorber and set it aside so that it does not interfere with the servicing. (See Front ABS Wheel-Speed Sensor Wiring Harness Installation Note.)
5. Remove the following parts:
   (1) Splash shield (See SPLASH SHIELD REMOVAL/INSTALLATION.)
   (2) Front under cover No.2 (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
   (3) Front under cover No.1 (See FRONT UNDER COVER No.1 REMOVAL/INSTALLATION.)
   (4) Insulator (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



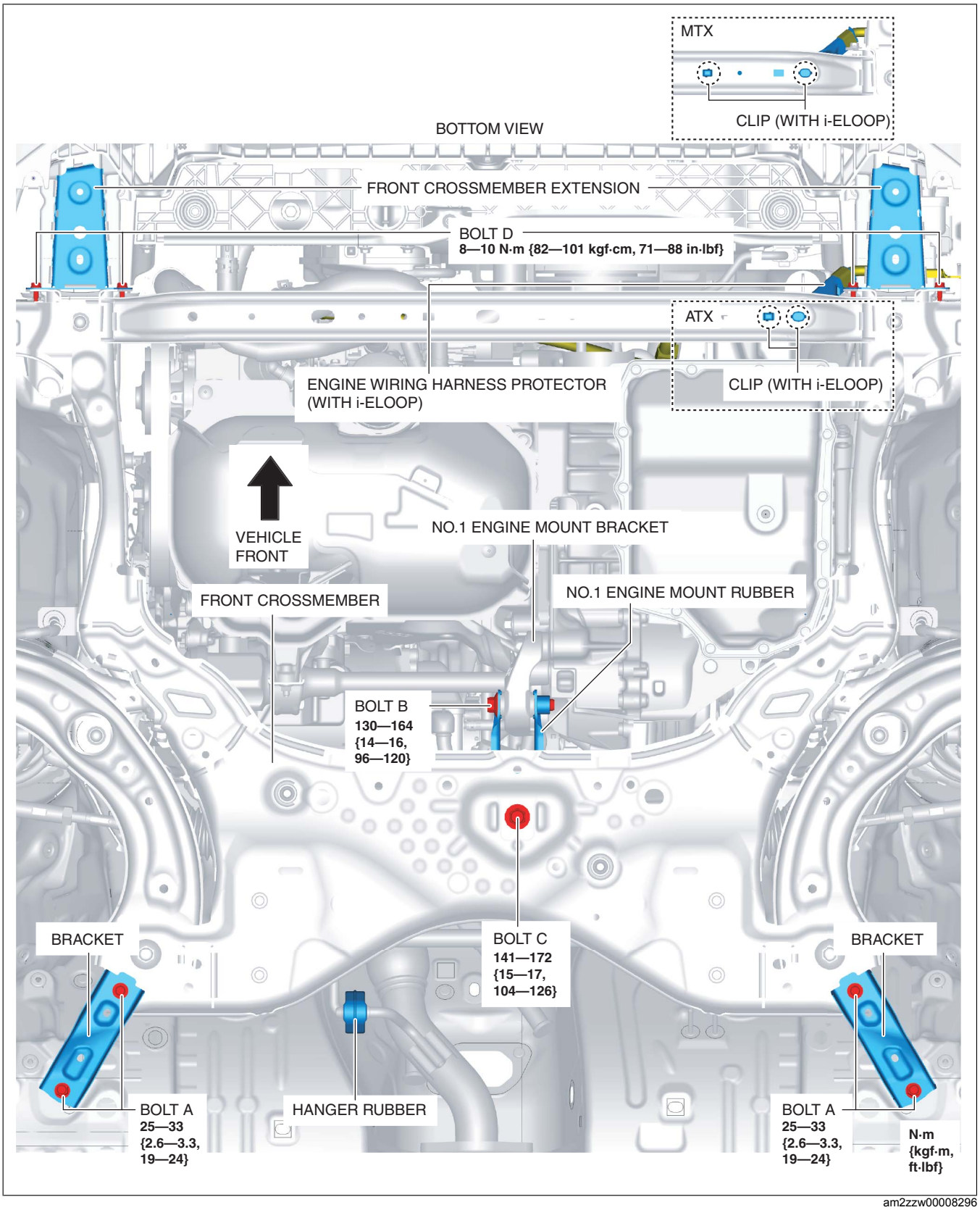
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6. Remove the screws and the fasteners shown in the figure and set the front mudguard aside.



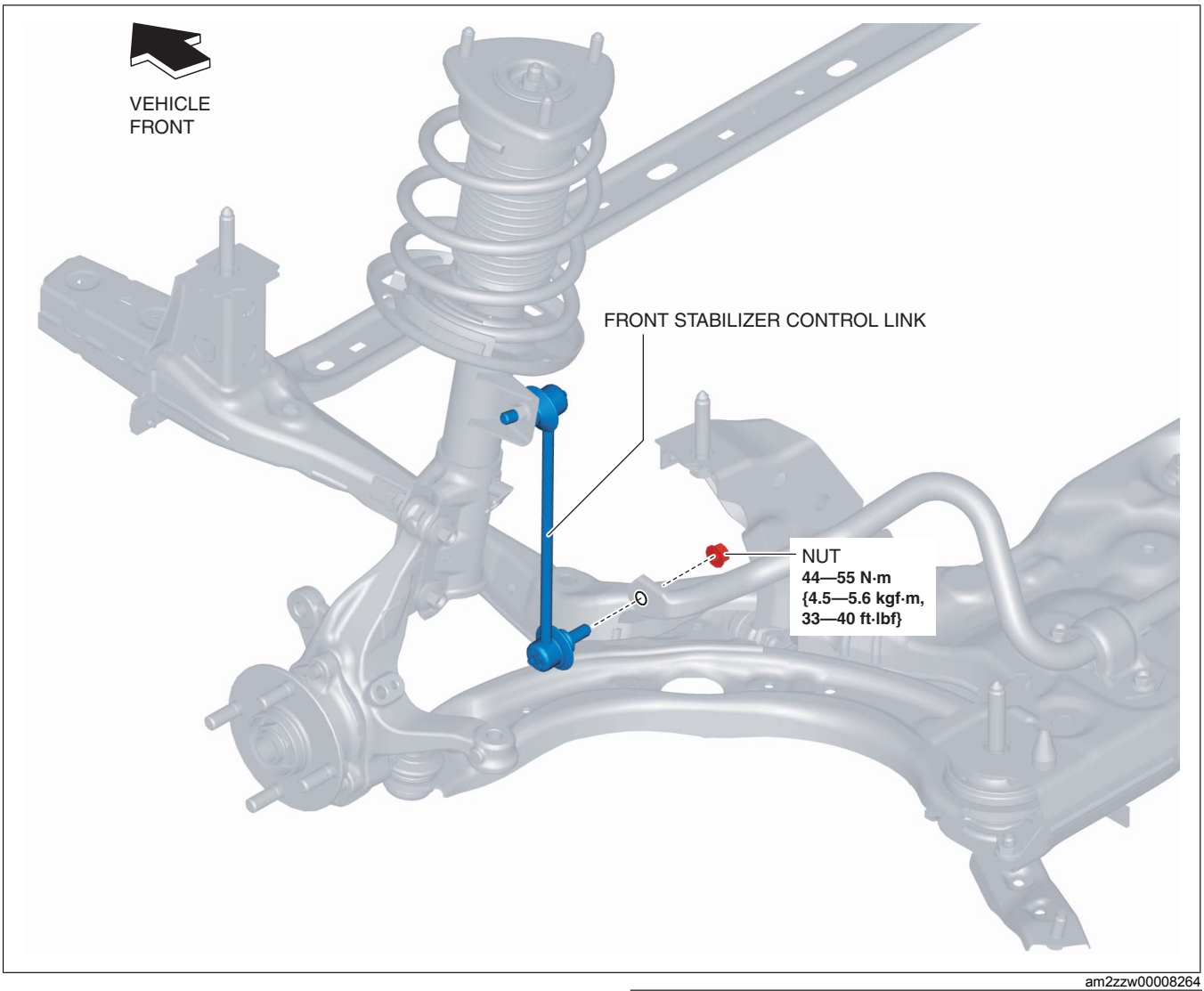
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7. Remove engine wiring harness protector clips from the front crossmember. (With i-ELOOP)



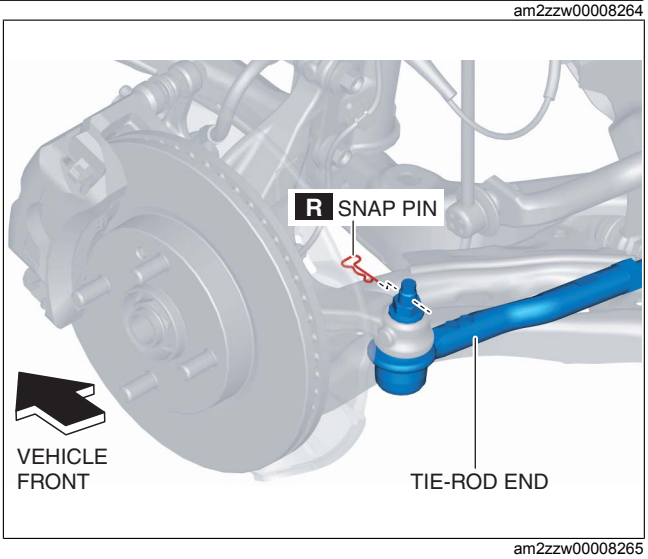
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8. Remove bolts A.
9. Remove the brackets.
10. Disconnect the hanger rubber from the front crossmember and set it aside.
11. Remove bolt B.
12. Remove bolt C.
13. Disengage the No.1 engine mount rubber and the No.1 engine mount bracket. (See No.1 Engine Mount Rubber and No.1 Engine Mount Bracket Installation Note.)
14. Disconnect the front stabilizer control link (front stabilizer side).



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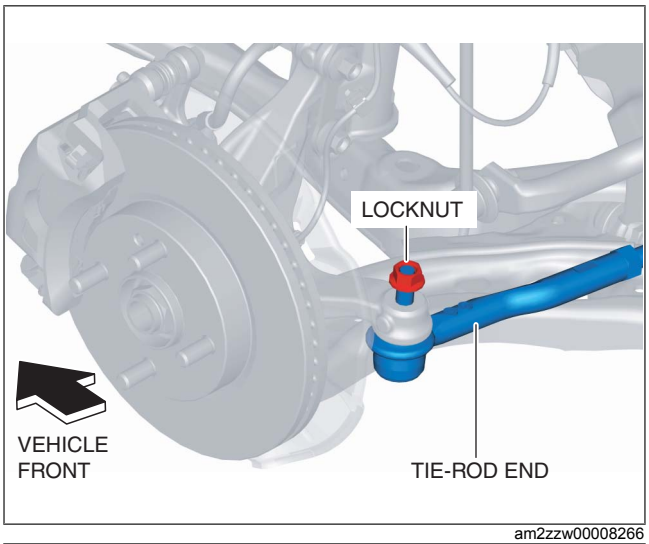
15. Remove the snap pin from the tie-rod end.



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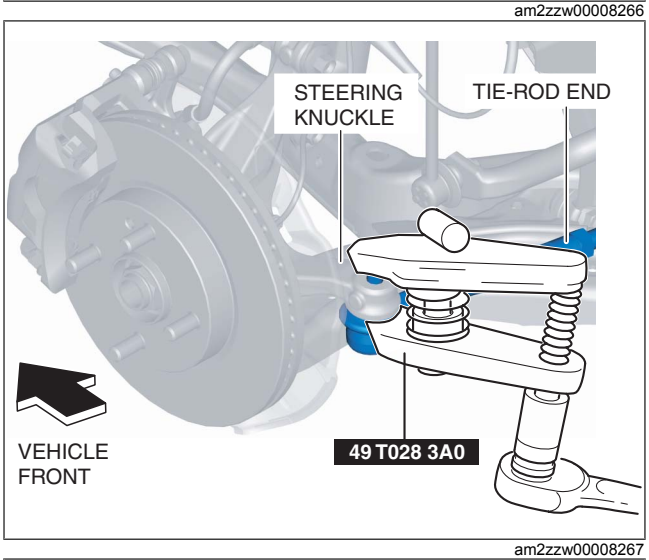
16. Loosen the locknut of the tie-rod end.

Tightening torque
47—59 N·m {4.8—6.0 kgf·m, 35—43 ft·lbf}



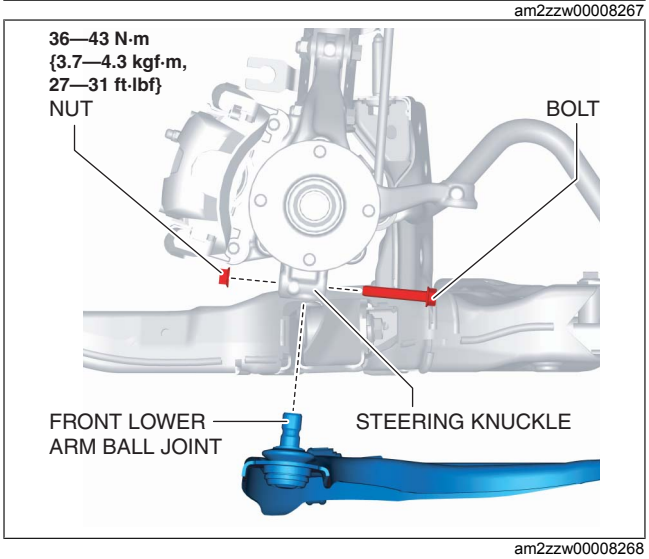
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17. Using the **SST**, disengage the tie-rod end from the steering knuckle.
18. Remove the locknut from the tie-rod end.
19. Disconnect the tie-rod end from the steering knuckle.



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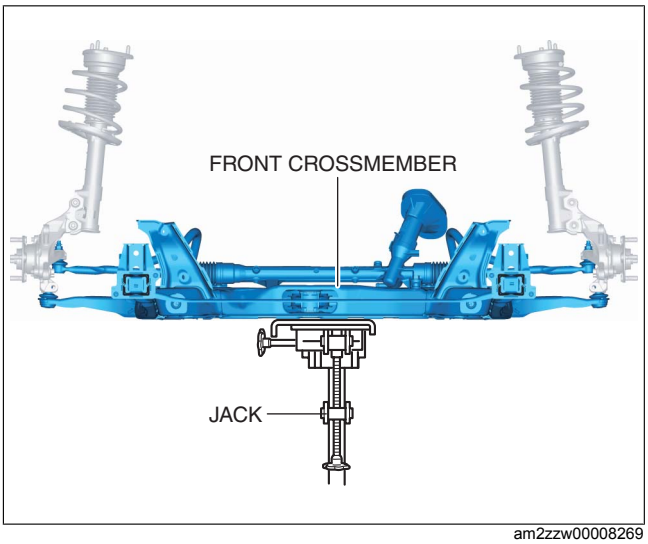
20. Remove the bolt and nut from the front lower arm ball joint.
21. Disconnect the front lower arm joint from the steering knuckle.



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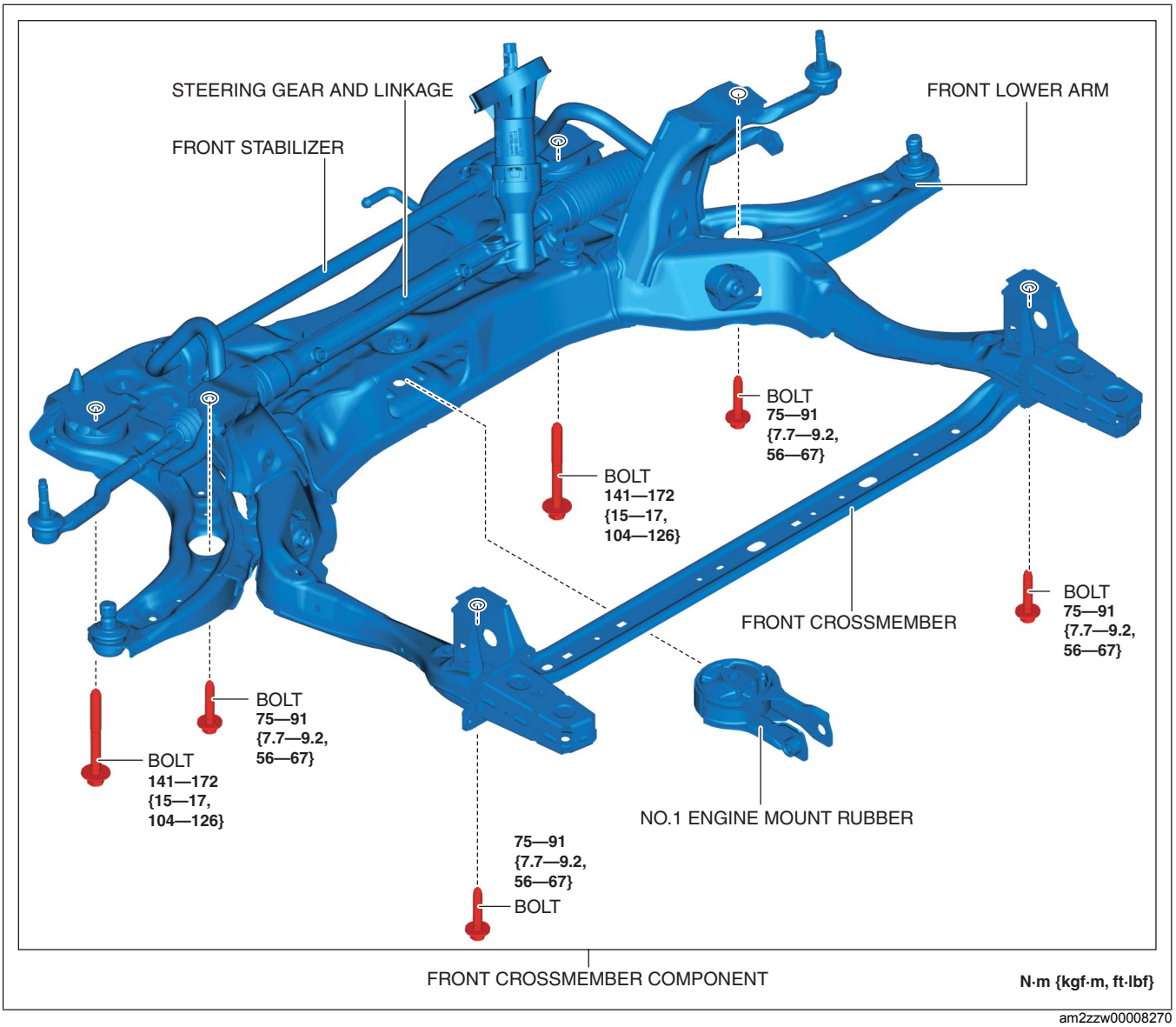
22. Support the front crossmember using a jack.

Warning
- **If the front crossmember falls off during front crossmember removal/installation, it may cause serious injury or death, or vehicle damage. When removing/installing the front crossmember, use a jack and verify that the crossmember is securely supported.**



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23. Remove the bolts.



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24. Lower the jack slowly and remove the front crossmember component. (See Front Crossmember Component Installation Note.)
25. Remove the following parts:
    (1) No.1 engine mount rubber
    (2) Front stabilizer (See FRONT STABILIZER REMOVAL.) (See FRONT STABILIZER INSTALLATION.)
    (3) Front lower arm (See FRONT LOWER ARM REMOVAL/INSTALLATION.)

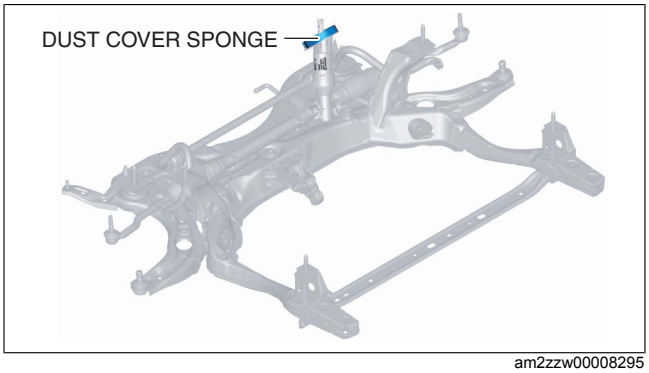
(4) Steering gear and linkage (See STEERING GEAR AND LINKAGE REMOVAL/INSTALLATION.)

26. Install in the reverse order of removal.
27. Inspect the wheel alignment and adjust it if necessary. (See FRONT WHEEL ALIGNMENT.)

Front Crossmember Component Installation Note

Caution

- **When installing the front crossmember component to the vehicle, assemble the dust cover to the body-side installation area correctly to prevent water or foreign matter from penetrating. Install the dust cover carefully so as not to pinch the dust cover sponge to the body-side installation area or put a fold line on the sponge by contact.**



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Front Lower Arm Ball Joint Installation Note

1. Install the front lower arm ball joint to the steering knuckle, insert the bolts in the same direction for the right and left, then tighten them to the specified torque.

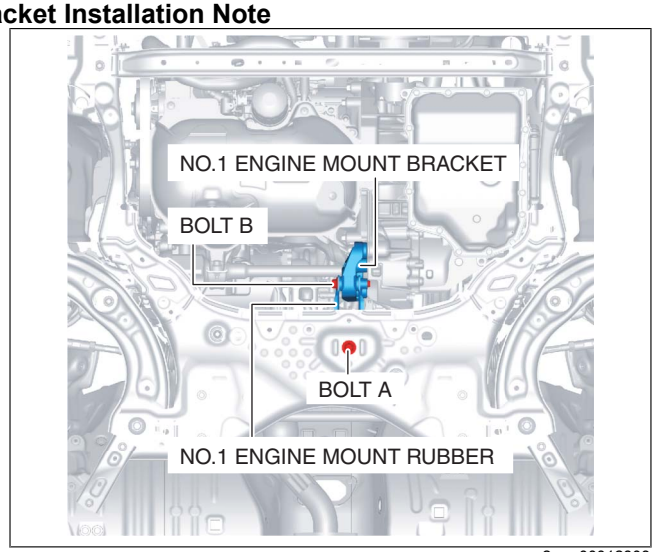
No.1 Engine Mount Rubber and No.1 Engine Mount Bracket Installation Note

1. Temporarily tighten bolt A shown in the figure.
2. Temporarily tighten bolt B shown in the figure.
3. Tighten bolt A.

   Tightening torque
   141—172 N·m {15—17 kgf·m, 104—126 ft·lbf}

4. Tighten bolt B.

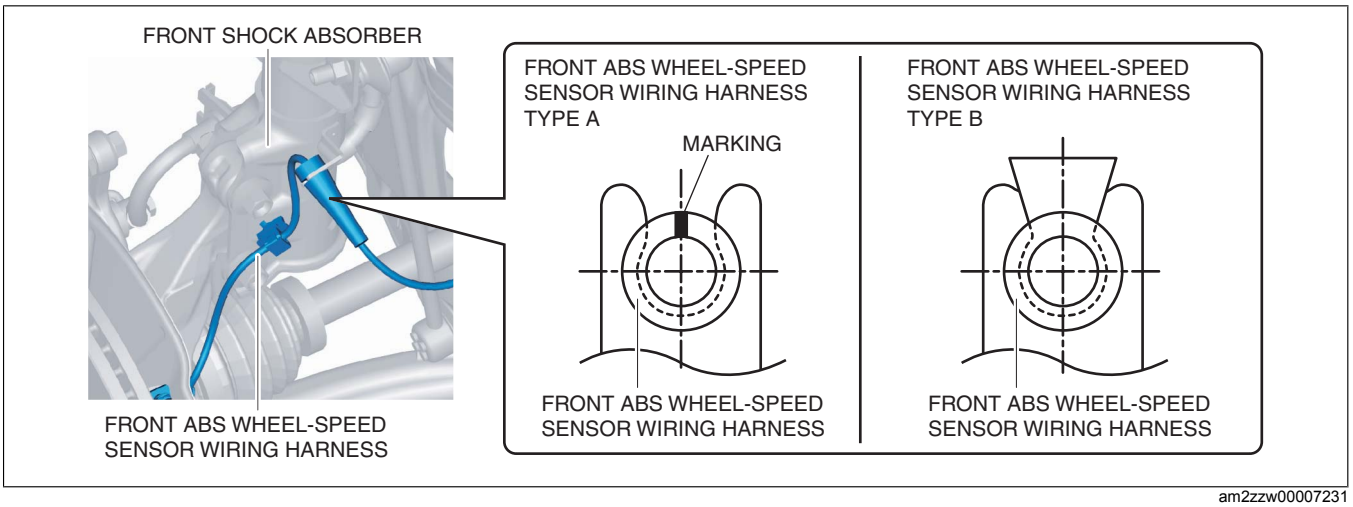
   Tightening torque
   130—164 N·m {14—16 kgf·m, 96—120 ft·lbf}



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Front ABS Wheel-Speed Sensor Wiring Harness Installation Note

1. Install the front ABS wheel-speed sensor wiring harness as shown in the figure.



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